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Short Answer Test

1) Given,

$E \to E+E \mid id$

$E \to E+E$

$E \to id$

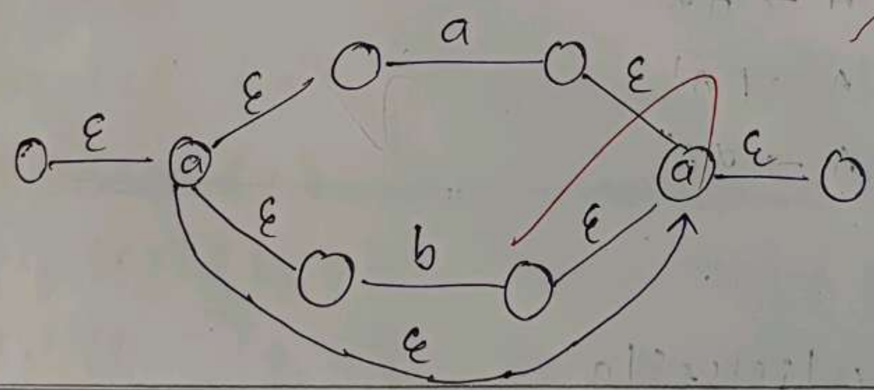
54%

$\frac{54}{100}$ ~~40~~

It is Ambiguous because left hand side & Right hand side have the same terminals

2) a) Given,

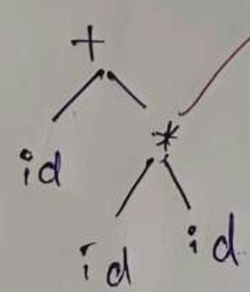
$a(a|b)^*a$



3) Given,

id + id * id

Parse Tree :-



id — +
|
id — *
|
id.

4) Given,

$A \to A\alpha \mid \beta$

Left recursion :- when LHS and RHS have the same Non-Terminal elements do left recursion

$A \to A\alpha \mid \beta$

$A \to \beta$

$A \to A\alpha$

$A \to \beta A'$

$A \to \alpha A'$

5) Given,

$S \to iEtS \mid iEtSeS \mid a$

6) $S \rightarrow aBDh$

$B \rightarrow cC$

$C \rightarrow bC / \varepsilon$

$D \rightarrow EF$

$E \rightarrow g / \varepsilon$

$F \rightarrow f / \varepsilon$

3

FIRST (S) : {a}
FIRST (B) : {c}
FIRST (C) : {b, ε}
FIRST (D) : {g, ε}
FIRST (E) : {g, ε}
FIRST (F) : {f, ε}

FOLLOW (S) : {$}
FOLLOW (B) : {g, f, h}
FOLLOW (C) : {g, f, h}
FOLLOW (D) : {h}
FOLLOW (E) : {f, h}
FOLLOW (F) : {h}

7) Given.

$S \rightarrow Sa | Sb | cd$

$S \rightarrow Sa$

$S \rightarrow Sb$

$S \rightarrow cd$

$S \rightarrow bS'$

$S' \rightarrow aS$

$S \rightarrow aS \mid bS' \mid \varepsilon$

It contains immediate left recrusion of

$S \rightarrow Sa \mid Sb \mid cd$

2

8) Given,

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

FIRST(E') : $\{+, \varepsilon\}$

2

9) Given,

$S \rightarrow (S) S \mid \varepsilon$

10) In shift reduce parsing we use

* shift
* Reduce
* Accept
* Error.

2

Shift :- when a element in the stack is less than expression do the shift transfer from RHS to LHS

Reduce :- when LHS & RHS Elements if LHS element is greater than RHS element then do Reduce

$S \rightarrow id + id$ $\quad$ $S \rightarrow S + S$

$S \rightarrow id$

Stack	expression	
S	$ id+id $	shift
$S \rightarrow id$	+id $	Reduce $S \rightarrow id$
$S \rightarrow S$	+id $	shift
$S \rightarrow S+$	id $	Reduce $S \rightarrow id$
$S \rightarrow S + id$	$	
$S \rightarrow S+S$	$	$S \rightarrow S+S$
$S \rightarrow S$	$	$S \rightarrow S$
S	$	Accept

11) Given,

Grammer : $S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC \mid d$

item : $[S' \rightarrow \cdot S]$

Closure :-

$S' \rightarrow S$

$S \rightarrow \cdot CC$

$S \rightarrow \cdot cC \mid \cdot d$

12) Given,

Non Terminal

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

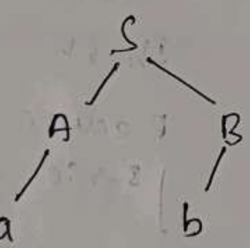
$S \rightarrow aSb \mid ab$

check aabb

FIRST (S) = {A}

FIRST (A) = {a}

FIRST (B) = {b}



∴ aabb is Not possible.

13) Given,

$S \rightarrow A$

$A \rightarrow aB \mid Ad$

$B \rightarrow b$

$C \rightarrow g.$

$A \rightarrow A'd$

$A \rightarrow dA'$

FIRST(S): {a}

FIRST(A): {a, d}

FIRST(B): {b}

FIRST(C): {g}

FIRST(A'): {d}

FOLLOW(S): {$}

FOLLOW(A): {b, $}

FOLLOW(B): {b, $}

FOLLOW(C): {g, b, $}

FOLLOW(A'): {b, $}

2

14) Code:

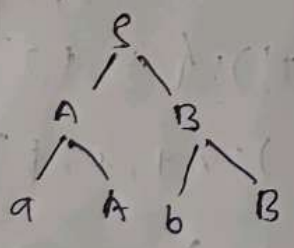
```
LDF   R1, R2
ADF   A1, R1, R2
STF   id, R1
```

1

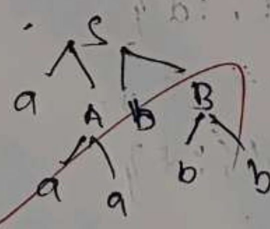
15) LMD & RMD

$S \rightarrow AB$

$A \rightarrow aA \mid a \quad B \rightarrow bB$



LMD



RMD

2

16) $S \rightarrow AB$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid C$

FOLLOW(S) : { $ }

FOLLOW(A) : { $, a }

FOLLOW(B) : { $, b, c }

FIRST(S) : { a }

FIRST(A) : { a, ε }

FIRST(B) : { b, c }

18) Given,

$E \to TE'$

$E' \to TE' | \epsilon$

$T \to FT'$

$T' \to *FT' | \epsilon$

$F \to (E) | id$

id + id * id

FIRST(E) : { (, id }

FIRST(E') : { (, id }

FIRST(T) : { (, id }

FIRST(T') : { *, ε }

FIRST(F) : { (, id }

FOLLOW(E) : { $ }

FOLLOW(E') : { $,) }

FOLLOW(T) : { +, $,) }

FOLLOW(T') : { +, $,) }

FOLLOW(F) : { +, *, $ }

Normilization	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE$		
E'		$E' \to TE'/\epsilon$				
T'	$T \to ET'$			$T \to ET'$		
T'			$T' \to *FT'/\epsilon$			
F	$F \to (E)/id$			$F \to (E)/id$		

Given

$E \rightarrow E+E$

$E \rightarrow E*E$ $\quad\quad$ $id_1 + id_2 * id_3$

$E \rightarrow (E)$

$E \rightarrow id$

Stack	input	shift/reduce
	$id_1+id_2*id_3\$$	shift
id_1	$+id_2*id_3\$$	Reduce $E \rightarrow id$
$E+$	$id_2*id_3\$$	Shift
$E+id_2$	$*id_3\$$	Reduce $E \rightarrow id$
$E+E$	$*id_3\$$	Reduce $E \rightarrow E+E$
E	$*id_3\$$	shift
$E*$	$id_3\$$	shift
$E*id_3$	$\$$	Reduce $E \rightarrow id$
$E*E$	$\$$	Reduce $E \rightarrow E*E$
E	$\$$	Accepted

9

20)
```
# include <stdio.h>
int main() {
    int a=10, b=20;
    int sum = a+b;
    printf("%d", sum);
    return 0;
}
```

a) Lexical Analysis

↓

Syntax Analysis

↓

Semtoric Analysis

↓

Intermediate Code Generator

↓

Code Optmization

↓

Code Generator

b) sum = a+b

<id> ; <=> <id;> <id3>

$t_1 = id_3$

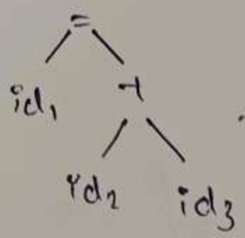
$t_2 = t_1 + id_2$

$t_3 = t_2$

$id_1 = t_3$

$t_1 = id_2 + id_3$

$id_1 = t_1$



c) Given,

sum = a + b

$S \rightarrow id = E$

$E \rightarrow E + T \,/\, T \Rightarrow E \rightarrow +TE'$

$E' \rightarrow TE'$

$T \rightarrow id \,/\, num$